DIPLOMA IN ELECTRICAL AND ELECTRONICS ENGINEERING • SEMESTER 5 • PROGRAM CORE

Synchronous Machines and FHP Motors

A full-width engineering handbook for Course Code 5031. This page covers alternator construction, EMF equation, armature reaction, voltage regulation methods, synchronizing, synchronous motor performance, V-curves, starting, hunting, synchronous condenser operation and fractional horsepower motors with diagrams, formulas, solved problems and exam answer guidance.

COURSE CODE
5031

COURSE TITLE
Synchronous Machines and FHP Motors

CREDITS
4

PERIODS
60

PERIODS / WEEK
4 (L:3 T:1 P:0)

COURSE CATEGORY
Program Core

COGNITIVE LEVEL
Applying + Understanding

BEST USE
Revision, numericals, drawings

□□□ short note □□□□. Alternator-□ EMF equation, phasor diagram, voltage regulation, synchronizing conditions □□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□ exam answer weak □□□□. FHP motor □□□□ motor selection, application, troubleshooting □□□□□□□□□□□□□ connect □□□□□□□□.

COURSE OVERVIEW

What 5031 expects you to actually know

The syllabus is built around four outcomes: alternator EMF and performance, voltage regulation and parallel operation, synchronous motor performance, and fractional horsepower motors. The official course has 60 periods with 4 credits and belongs to the Program Core area.

Objectives

- ✓ Understand construction, working principle and characteristics of synchronous machines and FHP motors.
- ✓ Understand parallel operation of alternators.
- ✓ Carry out testing of three-phase synchronous machines and single-phase motors.
- ✓ Select suitable machines based on applications.

Prerequisites

Topic	Course	Semester	Why it matters
Basic electrical engineering	Fundamentals of Electrical Circuits	3	AC quantities, power, impedance, phasors
DC Machines	DC Machines and Traction Motors	3	EMF, torque, losses, testing
Induction machines	Induction Machines	4	Rotating field, slip, single-phase motor basics

Course outcomes with engineering meaning

CO	Official outcome	Hours	Engineering meaning	Exam output
CO1	Develop the EMF equation of a synchronous generator and explain its performances.	15	You must connect construction, speed-frequency relation, winding factors, armature reaction and synchronous impedance.	Derivation, phasor diagrams, numericals.
CO2	Apply methods to pre-determine voltage regulation and illustrate parallel operation of synchronous generators.	15	You must use OC/SC/ZPF data, regulation methods, synchronizing conditions and load sharing logic.	Test procedure, calculation, synchronizing diagram.
CO3	Illustrate the performance of synchronous motors.	14	You must explain constant speed, excitation effect, V-curves, starting, hunting and synchronous condenser use.	V-curve, phasor, comparison, applications.
CO4	Summarize working and applications of different fractional horsepower motors.	14	You must identify motor type from application and explain working: split phase, capacitor, shaded pole, universal, stepper, servo and special motors.	Construction, working, comparison, selection.

How to study this subject

1. Draw every machine before writing theory. If your diagram is missing, marks will drop.
2. For alternator answers, always connect frequency, flux, turns and winding factors.

3. For regulation problems, write the method first: EMF, MMF or ZPF. Do not mix steps.
4. For synchronous motor, learn excitation effect using V and inverted V curves.
5. For FHP motors, study by application: fan, mixer, printer, servo drive, actuator, timer.

Brutally honest exam warning

Memorising definitions only will not work here. This subject is diagram-heavy and calculation-heavy. A passable answer needs diagram + principle + formula + phasor/waveform + application. The weakest answers usually fail because students confuse alternator and induction motor behaviour, or write single-phase motor names without starting method.

> Diagram □□□□□□□ machine subject □□□□□□□□□□□□ practical □□□□. Field-□ motor fault □□□□□□□□□□□□ selection □□□□□□□□□ construction + current path + application □□□□□□□□□□□□□□□.

REDESIGNED SYLLABUS MAP

From official syllabus to engineering roadmap

This converts the module table into a practical learning map: what to draw, what to calculate, what to explain, and where it is used.

CO1 • 15 HOURS

Synchronous Generator

Core topics: Working principle, production of sinusoidal EMF, stator, rotor, armature winding, damper winding, speed-frequency relation, pitch factor, distribution factor, alternator on load, armature reaction and synchronous impedance.

Draw	Construction diagram, salient/cylindrical rotor comparison, EMF derivation waveform, UPF/ZPF armature reaction diagrams, loaded alternator phasor.
Calculate	Ns/f/P relation, EMF equation, pitch/distribution factor effect, synchronous impedance and regulation basics.
Explain	Why armature is stationary, why rotor carries DC field, why terminal voltage changes on load.
Field use	DG alternator, power plant generator, brushless excitation, standby power panel.
Exam importance	Very high: derivation + phasor + numericals are repeated topics.

CO2 • 15 HOURS

Regulation and Parallel Operation

Core topics: Open circuit test, short circuit test, EMF method, MMF method, ZPF method, losses, efficiency, power developed, parallel operation, dark lamp, bright lamp, synchroscope, synchronizing current, power and torque definitions.

Draw	OCC/SCC, ZPF triangle, EMF and MMF phasor diagrams, power flow diagram, synchronizing lamps and synchroscope.
Calculate	Synchronous impedance, synchronous reactance, voltage regulation, efficiency and synchronizing quantities definitions.
Explain	Why alternators are paralleled, synchronizing conditions, load sharing logic.
Field use	DG synchronization panel, utility grid connection, power plant busbar operation.
Exam importance	Very high: regulation problems and synchronizing conditions are scoring areas.

CO3 • 14 HOURS

Synchronous Motor

Core topics: Synchronous motor construction, working principle, load effect at constant excitation, excitation effect at constant load, power developed equation, losses, power flow, V curves, inverted V curves, starting methods, hunting, comparison with induction motor, synchronous condenser.

Draw	Motor phasor diagram, V curve, inverted V curve, power flow, synchronous condenser vector diagram.
Calculate	Power developed equation, simple power angle problems, efficiency concept.
Explain	Why motor is not self-starting, how excitation changes power factor, how hunting is reduced.
Field use	Power factor correction, compressor drives, large pumps, constant speed drives.
Exam importance	High: V-curve and starting methods are frequent questions.

CO4 • 14 HOURS

FHP Motors

Core topics: Single-phase induction motors, double field revolving theory, split-phase, capacitor-start induction-run, capacitor-start capacitor-run, permanent capacitor, shaded pole, AC series, universal, repulsion, stepper, servo, reluctance, hysteresis, switched reluctance and applications.

Draw	Split-phase circuit, capacitor motor circuit, shaded pole pole-face, universal motor, stepper rotor/stator, servo block diagram.
Calculate	Simple speed/step angle/selection problems; mostly theory and applications.
Explain	Starting method, torque nature, direction reversal, application suitability and limitations.
Field use	Fans, blowers, pumps, mixers, washing machines, printers, scanners, actuators, small tools, control systems.

Exam importance	High for short notes, comparisons and application-selection questions.

CO1 • MODULE 1 • 15 HOURS

Construction, EMF equation, armature reaction and synchronous impedance

A synchronous generator, usually called an alternator, converts mechanical power into AC electrical power at a frequency locked with rotor speed. Its rotor carries a DC magnetic field and its stator has the three-phase armature winding. When the rotor is driven by a prime mover, flux cuts the stator conductors and induces sinusoidal EMF.

Textbook explanation

The stator is preferred for armature winding because the generated voltage and current are large. Keeping armature stationary gives better insulation, easier cooling and simpler connection to load through terminals instead of slip rings.

The rotor may be salient pole or cylindrical. Salient pole rotors are used for low-speed machines with many poles, such as hydro alternators. Cylindrical rotors are used for high-speed turbo alternators because they have better mechanical strength.

Pitch factor accounts for short-pitched coils. Distribution factor accounts for coils distributed in several slots. Both reduce the fundamental EMF slightly but improve waveform and reduce harmonics.

On load, terminal voltage changes due to armature resistance drop, leakage reactance drop and armature reaction. Armature reaction depends strongly on load power factor.

> Alternator-□ stator winding □□□ output □□□□□□□□□□□□□. Rotor DC field rotate □□□□□□□□□□□. Speed, poles, flux, turns, winding factors □□□□□□□ generated voltage □□□□□□□□□□□□□□□□□□.

Subtopic cards

Stationary armature

High voltage insulation and cooling become easier. The load current is taken from fixed terminals, so large current slip rings are avoided.

Rotor types

Salient pole has projected poles and large diameter. Cylindrical rotor is smooth, strong and suited for high speed.

Damper winding

Copper bars embedded in pole faces reduce hunting and help stable operation.

EMF equation

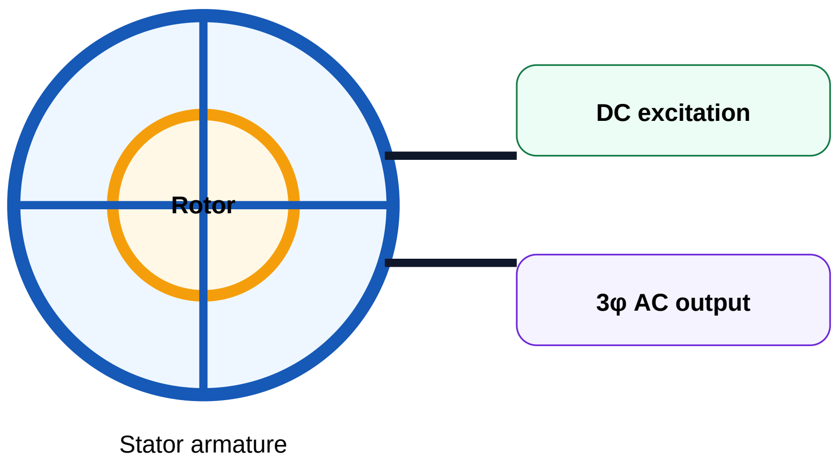
The generated phase EMF is proportional to frequency, flux per pole, series turns and winding factors.

Armature reaction

The armature current creates its own magnetic field and changes main field depending on power factor.

Synchronous impedance

Combined effect of armature resistance and synchronous reactance used for regulation prediction.



Synchronous generator construction idea

Important formulas / principles

f = PN/120

Ns = 120f/P

$$E_{ph} = 4.44 K_p K_d f \Phi T$$

$$Z_s = V_{oc} \text{ per phase} / I_{sc} \text{ per phase}$$

$$X_s = \sqrt{(Z_s^2 - R_a^2)}$$

Common mistakes:

- Forgetting Kp and Kd in EMF equation.
- Using line voltage directly instead of phase voltage in star connection.
- Saying armature reaction is always demagnetising. It depends on power factor.
- Drawing rotor/stator without labels.

Worked examples

Frequency calculation

For a 12-pole alternator at 500 rpm, f = PN/120 = 12×500/120 = 50 Hz.

Stationary armature reason

In high-voltage alternators, keeping armature stationary makes insulation and terminal connection safer and simpler.

Expected questions from this module

Q1 Explain construction and working principle of synchronous generator.

Q2 Compare salient pole and cylindrical rotor.

Q3 Derive EMF equation of an alternator.

Q4 Explain pitch factor and distribution factor.

Q5 Explain armature reaction at UPF, ZPF lagging and ZPF leading.

Q6 Explain synchronous impedance and draw phasor diagram.

Exam answer-writing format

1. Write definition and practical purpose.
2. Draw the required diagram with labels.

3. Explain construction and working in sequence.
4. Add formula / phasor / waveform where applicable.
5. Finish with applications, advantage or caution.

CO2 • MODULE 2 • 15 HOURS

Predetermination of voltage regulation and synchronizing of alternators

Voltage regulation tells how much terminal voltage changes from no-load to full-load. Since testing a large alternator at full load is costly, regulation is often predetermined using indirect tests. Parallel operation allows several alternators to share load safely and economically.

Textbook explanation

Open circuit test gives generated voltage versus field current at rated speed. Short circuit test gives short-circuit current versus field current. From these two tests, synchronous impedance can be obtained.

The EMF method is simple but pessimistic because it treats armature reaction as equivalent reactance. MMF method separates field current components. ZPF method gives better practical regulation using Potier triangle.

Parallel operation is required for reliability, load growth, maintenance flexibility and improved efficiency. Alternators cannot be connected randomly to a live busbar.

Synchronizing conditions are equal voltage, equal frequency, same phase sequence and zero phase angle at instant of closing. Failure causes circulating current and mechanical shock.

> Alternator parallel □□□□□□□□□□□□ voltage, frequency, phase sequence, phase angle match □□□□□□□□□□□. □□□□□□□□ synchronizing □□□□ fault, breaker damage, shaft stress □□□□□□□□□□□.

Subtopic cards

OCC test

Run alternator at rated speed with armature open. Increase field current and note open circuit voltage.

SCC test

Short armature through ammeters, run at rated speed, increase field until rated current flows.

EMF method

Find Zs and Xs; calculate generated EMF using phasor equation; regulation follows.

MMF method

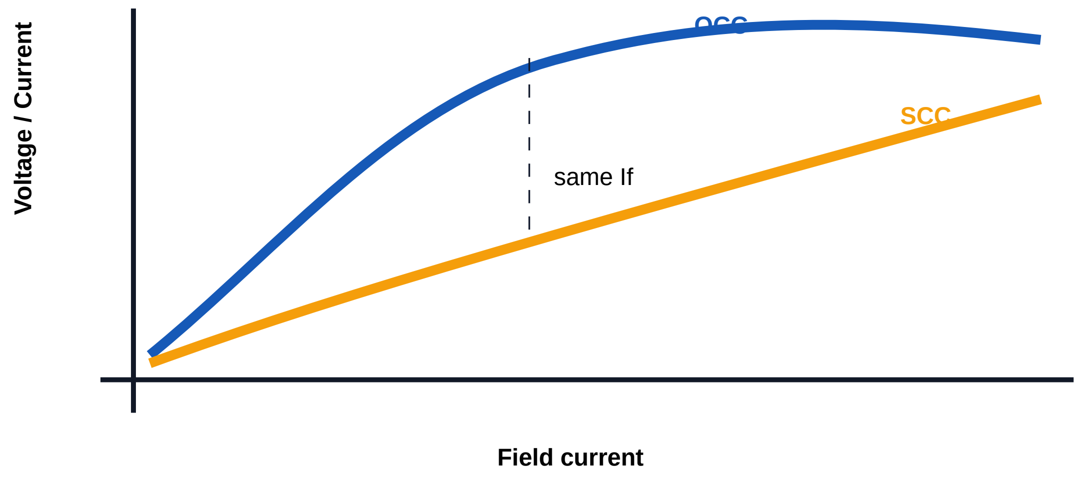
Find field ampere-turns for voltage and armature reaction separately and combine phasorially.

ZPF method

Uses zero power factor test and Potier triangle to separate leakage reactance and armature reaction.

Synchronizing

Dark lamp, bright lamp and synchroscope indicate correct closing instant.



Open circuit and short circuit characteristics

Important formulas / principles

% Regulation = (E0 - V) / V × 100

Zs = Voc / Isc at same field current

$Xs = \sqrt{(Zs^2 - Ra^2)}$

Efficiency η = Output / Input × 100

Pdev ≈ 3EV/Xs sinδ

Common mistakes:

- Not converting line voltage to phase voltage before regulation calculation.
- Closing breaker when lamps are not indicating same phase condition.
- Writing EMF, MMF and ZPF methods as the same method.
- Ignoring armature resistance if given.

Worked examples

Why parallel alternators?

One large load can be shared by multiple machines; one unit can be isolated for maintenance without full shutdown.

Why ZPF test?

It provides data to separate leakage reactance voltage drop and armature reaction effect more practically than EMF method.

Expected questions from this module

Q1 Explain OCC and SCC tests on an alternator.

Q2 Explain EMF method of voltage regulation.

Q3 Explain MMF method and ZPF method.

Q4 Define voltage regulation and efficiency of alternator.

Q5 Explain need and conditions for parallel operation.

Q6 Explain dark lamp, bright lamp and synchroscope methods.

Exam answer-writing format

1. Write definition and practical purpose.
2. Draw the required diagram with labels.
3. Explain construction and working in sequence.
4. Add formula / phasor / waveform where applicable.
5. Finish with applications, advantage or caution.

CO3 • MODULE 3 • 14 HOURS

Construction, performance, starting, hunting and synchronous condenser

A synchronous motor runs at exactly synchronous speed. Its rotor locks with the rotating magnetic field of the stator. Unlike induction motor, it has no slip during steady operation. The important feature is controllable power factor by changing excitation.

Textbook explanation

At constant excitation, increasing mechanical load increases torque angle and armature current. Speed remains practically constant until pull-out limit.

At constant load, changing field excitation changes armature current and power factor. Under-excited motor draws lagging current, normal excitation gives near unity power factor, and over-excited motor draws leading current.

The V curve is the plot of armature current against field current. The inverted V curve shows power factor against field current. These curves are very important for exam and practical PF correction.

Synchronous motor is not self-starting because average starting torque is zero. It can be started using damper winding, pony motor, variable frequency supply or by running as induction motor initially.

> Synchronous motor excitation □□□□□□□□□□□□ current □□□□□ □□□□□□□□□□ □□□□□□□ □□□□□□□□□ □□□□□□□□. Over excitation-□ leading current □□□□□□□□, □□□□□□□□□□ power factor correction-□ synchronous condenser □□□ □□□□□□□□□□□□□□□□.

Subtopic cards

Constant speed

Runs at Ns = 120f/P independent of load until pull-out.

Excitation effect

Field current controls power factor and armature current.

V curves

Armature current versus field current; minimum current near unity PF.

Starting

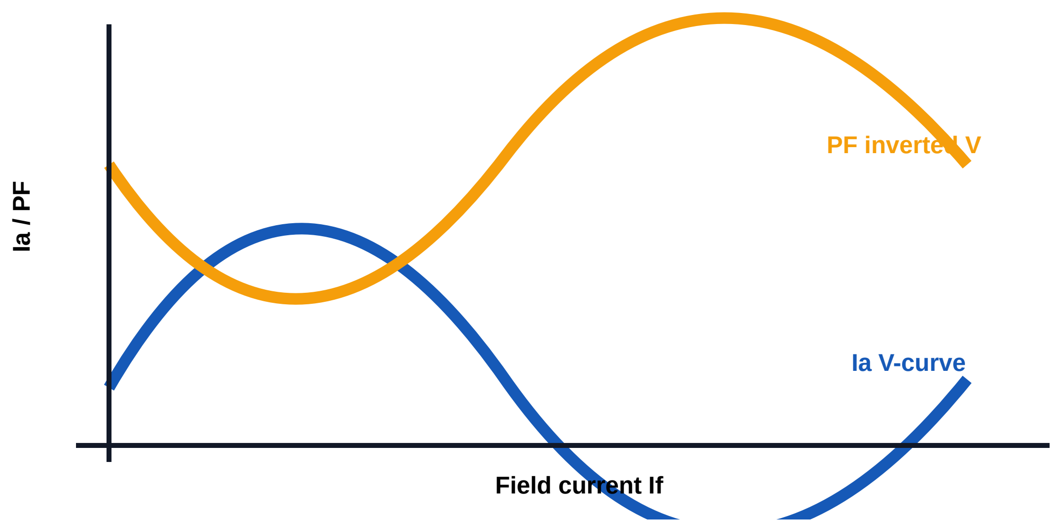
Damper winding/pony motor/VFD methods used because it is not self-starting.

Hunting

Oscillation of rotor around synchronous position due to sudden load change.

Synchronous condenser

Over-excited synchronous motor without mechanical load supplies leading kVAR.



Excitation effect in synchronous motor

Important formulas / principles

Ns = 120f/P

Pdev = 3VE/Xs sinδ

Torque T = Pω / ωs

For motor, under-excitation → lagging PF, over-excitation → leading PF

Common mistakes:

- Writing that synchronous motor has slip like induction motor.
- Saying it is self-starting. It is not self-starting.
- Drawing V curve without lagging/unity/leading regions.
- Not mentioning damper winding in hunting reduction.

Worked examples

PF correction

An over-excited synchronous motor draws leading current and can compensate lagging reactive power of induction motors.

Load change

When load increases, rotor angle δ increases to develop more torque while speed remains synchronous.

Expected questions from this module

Q1 Explain construction and working of synchronous motor.

Q2 Explain effect of load at constant excitation.

Q3 Explain effect of excitation at constant load with phasor approach.

Q4 Draw and explain V and inverted V curves.

Q5 Explain starting methods of synchronous motor.

Q6 Define hunting and methods to reduce it.

Q7 Compare synchronous motor and induction motor.

Q8 Explain synchronous condenser operation.

Exam answer-writing format

1. Write definition and practical purpose.
2. Draw the required diagram with labels.
3. Explain construction and working in sequence.

4. Add formula / phasor / waveform where applicable.
5. Finish with applications, advantage or caution.

CO4 • MODULE 4 • 14 HOURS

Single-phase induction, commutator and special purpose motors

Fractional horsepower motors are small motors rated below one horsepower. They may look simple, but correct selection depends on starting torque, running torque, speed, cost, noise, duty cycle, control requirement and maintenance.

Textbook explanation

A single-phase induction motor is not self-starting by itself because the pulsating field can be resolved into two equal opposite rotating fields. Starting winding or shading creates phase shift and starting torque.

Split-phase motors use starting and running windings. Capacitor motors improve phase shift and starting torque. Shaded pole motors are simple and cheap but have low starting torque and low efficiency.

Universal motors are commutator motors that can operate on AC or DC and provide high speed and high starting torque. They are common in mixers, drills and vacuum cleaners.

Stepper motors move in fixed angular steps for every input pulse. Servo motors use feedback for accurate position or speed control. Reluctance, hysteresis and switched reluctance motors are special-purpose motors with specific industrial uses.

> FHP motor selection-□ load torque, starting torque, cost, noise, duty cycle, speed control, maintenance □□□□□□□ □□□□□□□□. □□□□□ □□□□□□□ memorize □□□□□□□□ practical answer □□□□□□□□□□.

Subtopic cards

Double field revolving theory

Pulsating single-phase field is equivalent to two opposite rotating fields. Net starting torque is zero.

Split-phase motor

Auxiliary winding with high resistance creates phase shift for starting.

Capacitor motors

Capacitor improves starting torque and PF; types include capacitor-start and permanent capacitor.

Shaded pole motor

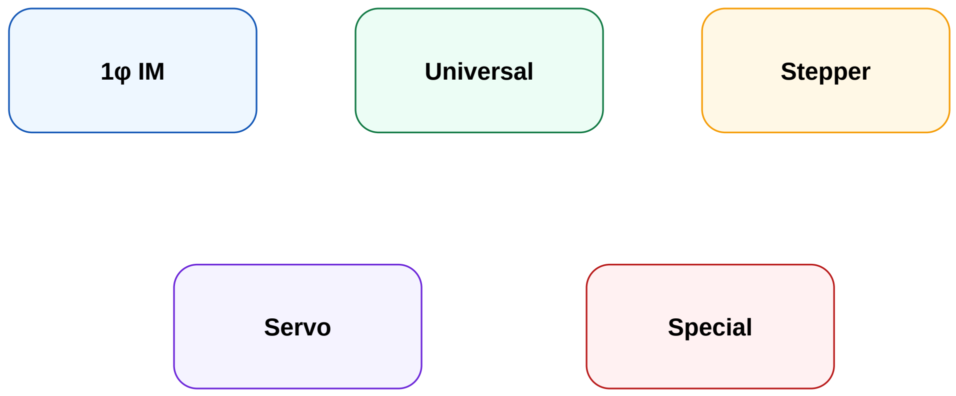
Copper shading ring creates weak rotating field; used in small fans and blowers.

Universal motor

Series commutator motor for AC/DC with high speed and starting torque.

Stepper and servo

Stepper works by pulses; servo uses feedback for closed-loop control.



FHP motor family

Important formulas / principles

Step angle β = 360° / (number of rotor teeth × number of phases) for common simplified stepper relation

Speed of stepper = step angle × pulse rate / 360 revolutions per second

FHP means motor rating less than 1 HP

Common mistakes:

- Saying single-phase induction motor is naturally self-starting.
- Selecting shaded pole motor for high starting torque load.
- Confusing universal motor with ordinary induction motor.
- Ignoring feedback in servo motor explanation.

Worked examples

Motor selection

Ceiling fan: permanent capacitor motor. Mixer grinder: universal motor. Printer paper feed: stepper motor. Position control: servo motor.

Shaded pole limitation

It is cheap and quiet but has low efficiency and low starting torque, so it is used only for very small loads.

Expected questions from this module

Q1 Explain double field revolving theory.

Q2 Explain split-phase induction motor.

Q3 Explain capacitor-start and capacitor-run motors.

Q4 Explain shaded pole motor with diagram.

Q5 Explain universal motor and repulsion motor.

Q6 Explain stepper motor types and applications.

Q7 Explain AC and DC servomotors.

Q8 List applications of FHP motors.

Exam answer-writing format

1. Write definition and practical purpose.
2. Draw the required diagram with labels.
3. Explain construction and working in sequence.

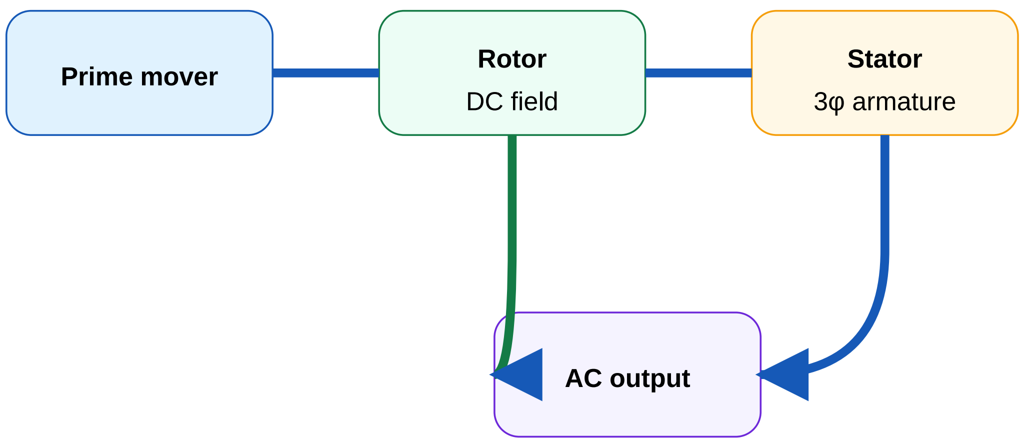
4 Add formula / phasor / waveform where applicable.

5 Finish with applications, advantage or caution.

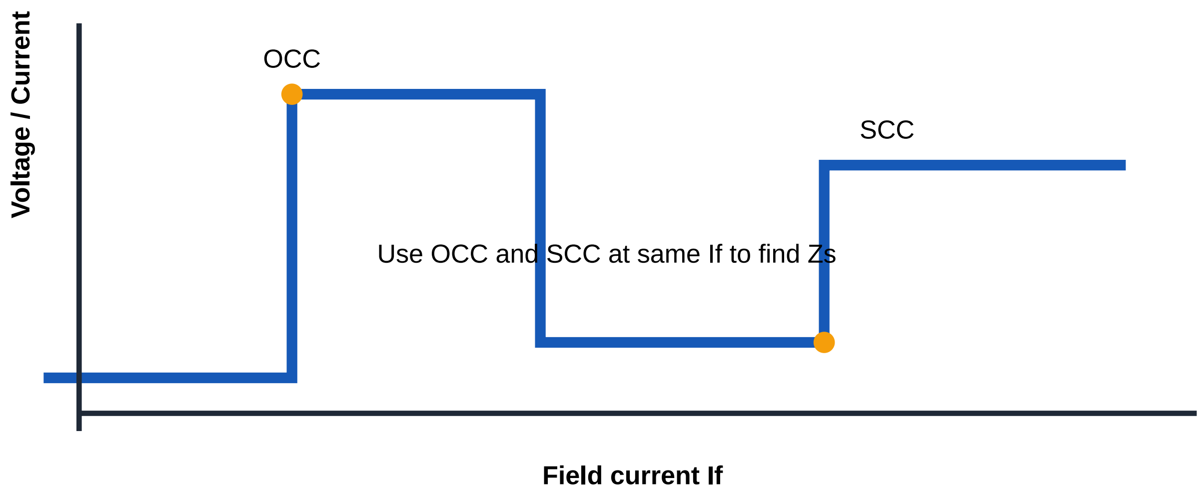
ILLUSTRATION LAB

Drawings you should practise until they become automatic

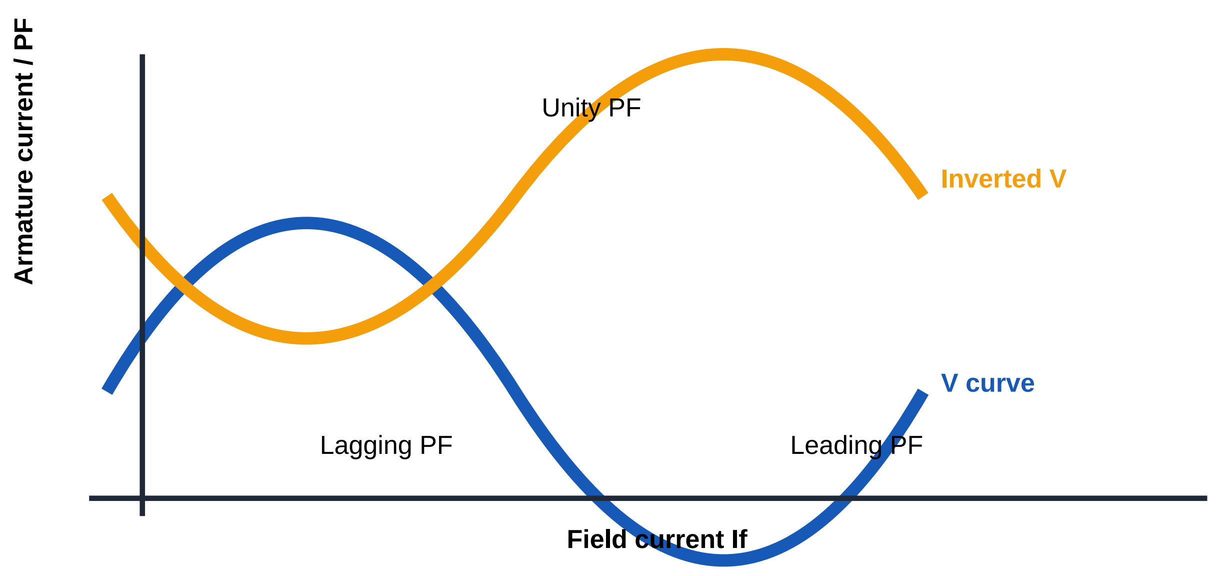
Use these as drawing checklists. Redraw them in your notebook because the exam often rewards a clean diagram even before the full explanation.



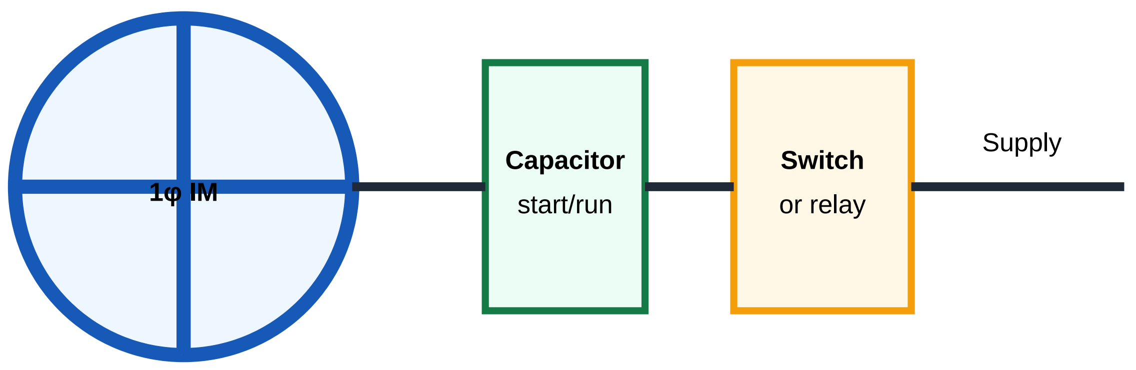
Alternator block diagram



OCC/SCC idea for synchronous impedance



Synchronous motor V and inverted V curves



Single-phase induction motor starting concept

FORMULA BANK

Formula, variable meaning, unit and direct use

Synchronous speed

$$N_s = 120f / P$$

Ns in rpm, f in Hz, P number of poles. Use to find alternator or motor speed.

Generated frequency

$$f = PN / 120$$

P poles and N rpm. Used in alternator speed-frequency problems.

Alternator EMF equation

$$E_{ph} = 4.44\, K_p\, K_d\, f\, \Phi\, T$$

Eph phase EMF in volts; Kp pitch factor; Kd distribution factor; Φ flux/pole in Wb; T series turns/phase.

Line voltage - star

$$V_L = \sqrt{3}\, V_{ph}$$

Use this conversion before alternator regulation calculation.

Line voltage - delta

$$V_L = V_{ph}$$

Delta line voltage equals phase voltage.

Synchronous impedance

$$Z_s = V_{oc} / I_{sc}$$

Use Voc per phase and Isc per phase at same field current.

Synchronous reactance

$$X_s = \sqrt{(Z_s^2 - R_a^2)}$$

Ra is armature resistance per phase.

Voltage regulation

$$\%Reg = (E_0 - V) / V \times 100$$

Positive for voltage drop; can be negative at leading PF.

Alternator efficiency

$$\eta = \text{Output} / \text{Input} \times 100$$

Input = output + losses; output is usually $\sqrt{3}\ V_L I_L \cos\varphi$ for 3φ.

3-phase power

$$P = \sqrt{3}\ V_L I_L \cos\varphi$$

Use for output power of alternator or motor input where line values are given.

Power developed

$$P_{dev} \approx 3EV/X_s \sin\delta$$

Cylindrical rotor approximate equation, ignoring resistance.

Torque

$$T = P / \omega_s$$

$\omega_s = 2\pi N_s/60$ rad/s.

Stepper speed

$$n = \beta \times \text{pulse rate} / 360$$

β step angle in degrees, n rev/s.

Stepper steps per revolution

$$\text{Steps} = 360 / \beta$$

Useful for stepper motor positioning problems.

SOLVED PROBLEM LAB

Exam-style calculations with steps, not just final answers

SOLVED 1

A 12-pole alternator runs at 500 rpm. Find generated frequency.

Given	P = 12, N = 500 rpm
Formula	f = PN / 120
Substitution	f = 12 × 500 / 120

Calculation	$f = 50$ Hz
Final answer	**50 Hz**
Exam tip	Always use mechanical rpm, not electrical speed.

SOLVED 2

Find phase EMF of an alternator if Kp = 0.96, Kd = 0.955, f = 50 Hz, Φ = 0.04 Wb and T = 120 turns/phase.

Given	$K_p=0.96$, $K_d=0.955$, $f=50$ Hz, $\Phi=0.04$ Wb, $T=120$
Formula	$E_{ph} = 4.44\, K_p K_d f \Phi T$
Substitution	$E_{ph} = 4.44 \times 0.96 \times 0.955 \times 50 \times 0.04 \times 120$
Calculation	$E_{ph} \approx 976.9$ V
Final answer	**Approximately 977 V per phase**
Exam tip	Do not forget both winding factors.

SOLVED 3

An alternator has Voc = 460 V per phase and Isc = 100 A at same field current. Armature resistance is 0.6 Ω/phase. Find Zs and Xs.

Given	$V_{oc}=460$ V/phase, $I_{sc}=100$ A, $R_a=0.6\ \Omega$
Formula	$Z_s=V_{oc}/I_{sc}$; $X_s=\sqrt{Z_s^2-R_a^2}$
Substitution	$Z_s=460/100=4.6\ \Omega$; $X_s=\sqrt{4.6^2-0.6^2}$
Calculation	$X_s=\sqrt{21.16-0.36}=\sqrt{20.8}=4.56\ \Omega$
Final answer	**$Z_s = 4.6\ \Omega$, $X_s \approx 4.56\ \Omega$**
Exam tip	Use phase voltage in Zs calculation.

SOLVED 4

A 3-phase alternator delivers 50 A at 230 V per phase, 0.8 lagging PF. Ra=0.4 Ω and Xs=3 Ω. Find approximate generated EMF per phase by EMF method.

Given	$V=230$ V, $I=50$ A, $\cos\varphi=0.8$ lag, $\sin\varphi=0.6$, $R_a=0.4\ \Omega$, $X_s=3\ \Omega$
Formula	For lag PF: $E = \sqrt{(V\cos\varphi + IR_a)^2 + (V\sin\varphi + IX_s)^2}$
Substitution	$E = \sqrt{(230\times0.8 + 50\times0.4)^2 + (230\times0.6 + 50\times3)^2}$
Calculation	$E = \sqrt{(184+20)^2 + (138+150)^2} = \sqrt{204^2 + 288^2} \approx 352.8$ V
Final answer	**$E \approx 353$ V per phase**
Exam tip	For lagging PF, reactive drop is added.

SOLVED 5

Using the previous problem, find percentage voltage regulation.

Given	E0=352.8 V, V=230 V
Formula	%Reg = (E0 - V)/V × 100
Substitution	%Reg = (352.8 - 230)/230 × 100
Calculation	%Reg ≈ 53.4%
Final answer	**Regulation ≈ 53.4%**
Exam tip	State positive regulation for lagging load.

SOLVED 6

A 6-pole synchronous motor is supplied from 50 Hz. Find synchronous speed.

Given	P=6, f=50 Hz
Formula	Ns=120f/P
Substitution	Ns=120×50/6
Calculation	Ns=1000 rpm
Final answer	**1000 rpm**
Exam tip	Synchronous motor speed is fixed by supply frequency and poles.

SOLVED 7

A stepper motor has step angle 1.8°. How many steps are required for one revolution?

Given	Step angle β=1.8°
Formula	Steps=360/β
Substitution	Steps=360/1.8
Calculation	Steps=200
Final answer	**200 steps/revolution**
Exam tip	Stepper positioning questions are easy if step angle is known.

SOLVED 8

A stepper motor with 7.5° step angle receives 120 pulses per second. Find speed in rpm.

Given	β=7.5°, pulse rate=120 pulses/s
Formula	rev/s = β × pulses/s / 360
Substitution	rev/s = 7.5×120/360 = 2.5
Calculation	rpm = 2.5×60 = 150 rpm

Final answer	**150 rpm**
Exam tip	Convert revolutions per second to rpm at the end.

PRACTICAL / INDUSTRIAL APPLICATION

Where this subject appears in real electrical engineering

Power generation

Alternators are used in thermal, hydro, diesel and gas plants. Regulation, excitation, synchronizing and load sharing are daily operating concerns.

Electrical panels

AMF panels, DG synchronization panels and power factor correction systems use synchronizing logic, voltage/frequency sensing, breaker interlocks and protection relays.

Industrial drives

Synchronous motors are used for constant-speed high-power applications and power factor correction. Excitation failure and hunting protection matter in real plants.

FHP appliances

Fans, blowers, mixers, drills, printers, scanners, timers, actuators and office equipment use split-phase, capacitor, shaded-pole, universal, stepper and servo motors.

Testing and troubleshooting

Open circuit, short circuit, resistance measurement, insulation test, capacitor test, winding continuity and abnormal noise/vibration checks are practical requirements.

Maintenance caution

Never parallel alternators without voltage, frequency, phase sequence and phase angle checks. Wrong synchronizing can create severe shaft torque, breaker damage and busbar fault.

QUICK REVISION

Last-day memory sheet

One-line memory points

- ✓ Alternator frequency: f = PN / 120.
- ✓ Generated EMF depends on flux, frequency, turns, pitch factor and distribution factor.
- ✓ Lagging load causes demagnetising armature reaction; leading load can be magnetising.

- ✓ High synchronous impedance means poor voltage regulation.
- ✓ Alternators must be synchronized only when voltage, frequency, phase sequence and phase angle match.
- ✓ Synchronous motor is not self-starting.
- ✓ Over-excited synchronous motor draws leading current and can improve power factor.
- ✓ Single-phase induction motor needs starting arrangement because starting torque is zero theoretically.

Diagram checklist

- ✓ Alternator construction: stator, rotor, slip rings/brushless excitation, damper winding.
- ✓ EMF derivation: show sinusoidal generated EMF and winding factor.
- ✓ Armature reaction: UPF, lagging ZPF, leading ZPF sketches.
- ✓ Synchronous impedance: OCC, SCC and phasor diagram.
- ✓ Parallel operation: dark lamp / bright lamp / synchroscope.
- ✓ Synchronous motor: V-curves, power flow, synchronous condenser vector.
- ✓ FHP motors: split phase, capacitor, shaded pole, universal, stepper, servo loop.

Common exam traps

- Writing $E = 4.44\ f\ \Phi\ T$ but forgetting Kp and Kd.
- Using line voltage in phase voltage formula for star-connected alternator.
- Not stating whether regulation is positive or negative at leading power factor.
- Mixing alternator power angle equation with induction motor slip formula.
- Writing stepper motor as a normal DC motor. It is position-controlled by pulses.

Answer-writing format

1. Start with definition and use.
2. Draw neat labelled diagram.
3. Explain construction before working.
4. Write formula with variable meanings.
5. Add practical application and one caution.

EXPECTED QUESTIONS

Module-wise short, descriptive, numerical and diagram questions

Module 1: Synchronous Generator

Short answer

- Explain construction and working principle of synchronous generator.
- Compare salient pole and cylindrical rotor.
- Derive EMF equation of an alternator.

Descriptive

- Explain pitch factor and distribution factor.
- Explain armature reaction at UPF, ZPF lagging and ZPF leading.
- Explain synchronous impedance and draw phasor diagram.

Diagram based

Construction diagram, salient/cylindrical rotor comparison, EMF derivation waveform, UPF/ZPF armature reaction diagrams, loaded alternator phasor.

Application based

DG alternator, power plant generator, brushless excitation, standby power panel.

Module 2: Regulation and Parallel Operation

Short answer

- Explain OCC and SCC tests on an alternator.
- Explain EMF method of voltage regulation.
- Explain MMF method and ZPF method.

Descriptive

- Define voltage regulation and efficiency of alternator.
- Explain need and conditions for parallel operation.

- Explain dark lamp, bright lamp and synchroscope methods.

Diagram based

OCC/SCC, ZPF triangle, EMF and MMF phasor diagrams, power flow diagram, synchronizing lamps and synchroscope.

Application based

DG synchronization panel, utility grid connection, power plant busbar operation.

Module 3: Synchronous Motor

Short answer

- Explain construction and working of synchronous motor.
- Explain effect of load at constant excitation.
- Explain effect of excitation at constant load with phasor approach.

Descriptive

- Draw and explain V and inverted V curves.
- Explain starting methods of synchronous motor.
- Define hunting and methods to reduce it.

Diagram based

Motor phasor diagram, V curve, inverted V curve, power flow, synchronous condenser vector diagram.

Application based

Power factor correction, compressor drives, large pumps, constant speed drives.

Module 4: FHP Motors

Short answer

- Explain double field revolving theory.
- Explain split-phase induction motor.
- Explain capacitor-start and capacitor-run motors.

Descriptive

- Explain shaded pole motor with diagram.
- Explain universal motor and repulsion motor.
- Explain stepper motor types and applications.

Diagram based

Split-phase circuit, capacitor motor circuit, shaded pole pole-face, universal motor, stepper rotor/stator, servo block diagram.

Application based

Fans, blowers, pumps, mixers, washing machines, printers, scanners, actuators, small tools, control systems.

MODEL QUESTION PAPER

Sample diploma examination pattern with marks distribution

This is an original model paper following typical diploma examination ordering. It does not copy official questions.

Course Code: 5031
Course Title: Synchronous Machines and FHP Motors
Time: 3 Hours Maximum Marks: 100

PART A - Short Answer Questions (10 × 2 = 20 marks)
1. Define pitch factor and distribution factor.
2. Write the relation between rotor speed, frequency and poles of an alternator.
3. State two advantages of stationary armature in an alternator.
4. Define voltage regulation of an alternator.

5. Name any two methods used to predetermine alternator voltage regulation.
6. State the conditions for parallel operation of alternators.
7. Why is a synchronous motor not self-starting?
8. What is hunting in a synchronous motor?
9. Name any two single-phase induction motor types.
10. State two applications of stepper motor.

PART B - Analytical / Descriptive Questions (8 × 5 = 40 marks)
11. Explain construction and working principle of a three-phase synchronous generator.
12. Derive the EMF equation of an alternator and define all symbols.
13. Explain armature reaction at UPF, ZPF lagging and ZPF leading loads with sketches.
14. Explain open circuit and short circuit tests and show how synchronous impedance is obtained.
15. Explain EMF method for voltage regulation with phasor diagram.
16. Explain dark lamp and synchroscope methods of alternator synchronizing.
17. Explain V-curves and inverted V-curves of a synchronous motor.
18. Explain construction and working of capacitor-start capacitor-run motor.

PART C - Long Answer / Numerical / Application Questions (4 × 10 = 40 marks)
19. A 3-phase, 16-pole alternator runs at 375 rpm. Find the frequency. If $\Phi = 0.05$ Wb, T = 80 turns/phase, Kp = 0.96 and Kd = 0.955, find generated EMF per phase.
20. An alternator has phase voltage 230 V, armature current 50 A, Ra = 0.4 Ω and Xs = 3 Ω. Find generated EMF and percentage regulation at 0.8 lagging power factor using EMF method.
21. Explain synchronous motor starting methods, effect of excitation and synchronous condenser operation with diagrams.
22. Compare split-phase, capacitor-start, shaded-pole, universal, stepper and servo motors. Select suitable motors for ceiling fan, mixer, printer paper feed, small blower and position control system with reason.

ANSWER KEY / WRITING GUIDE

Complete answer points, diagram requirement and mistakes to avoid

Answer 1

Pitch factor is ratio of EMF of short-pitched coil to full-pitched coil. Distribution factor is ratio of EMF of distributed winding to concentrated winding. Draw coil pitch sketch if asked for more marks.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 2

f = PN/120 and Ns = 120f/P. Mention P is number of poles and N is rpm.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 3

Stationary armature gives better insulation, easier cooling, fixed load terminals and avoids heavy-current slip rings.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 4

Voltage regulation is percentage change in terminal voltage from no-load to full-load at given PF and speed: $(E0 - V)/V \times 100$.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 5

Predetermination methods: EMF/synchronous impedance method, MMF/ampere-turn method and ZPF/Potier method.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 6

Parallel conditions: same voltage, same frequency, same phase sequence and zero phase angle at closing instant. Draw lamp/synchroscope diagram.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 7

A synchronous motor is not self-starting because average starting torque over one cycle is zero at standstill. Mention damper winding or VFD starting.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 8

Hunting is oscillation of rotor about its synchronous position due to sudden load or supply disturbance. Damper winding reduces it.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 9

Single-phase motor types include split-phase, capacitor-start, capacitor-start capacitor-run, permanent capacitor and shaded-pole motor.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 10

Stepper motors are used in printers, CNC positioning, robotics, scanners, disk drives and instrument pointers.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 11

For alternator construction: draw stator armature, rotor field, exciter, shaft and terminals. Explain stationary armature and rotor types.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 12

For EMF equation: start from average EMF per conductor, form factor 1.11 and include Kp and Kd. Finish with $E_{ph} = 4.44 K_p K_d f \Phi T$.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 13

For armature reaction: draw three cases. UPF mainly cross-magnetising, ZPF lagging demagnetising, ZPF leading magnetising.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 14

For regulation tests: draw OCC and SCC. Explain same field current method and calculate Zs and Xs.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 15

For synchronizing: draw busbar, incoming alternator and lamps/synchroscope. State conditions before closing breaker.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 16

For V-curves: draw Ia vs If and PF vs If. Mark lagging, unity and leading regions. Explain PF correction.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 17

For capacitor motor: draw main winding, auxiliary winding, capacitor and switch if applicable. Explain phase shift and starting torque.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

Answer 18

For numerical answers: use phase quantities, write formula, substitute values, show unit and round properly.

Do not miss: clean diagram or formula where applicable, unit for numerical answer, and one practical point.

REFERENCES

Textbooks and online resources from syllabus

Text / Reference books

- B. L. Theraja, Electrical Technology Vol. II, S. Chand & Co. Use for alternator and motor basics with simple diagrams.
- J. B. Gupta, Theory and Performance of Electrical Machines, S. K. Kataria & Sons. Use for regulation, phasors and machine performance.
- Prithwiraj Purkait and Indrayudh Bandyopadhyay, Electrical Machines, Oxford University Press. Use for deeper conceptual explanation.
- S. K. Sahdev, Electrical Machines, Cambridge University Press. Use for exam-style theory and practice problems.

Online resources

- NPTEL Electrical Engineering: use for lecture-level synchronous machine explanation.
- VSSUT Electrical Engineering lecture notes: use for supplementary machine notes.
- Electrical4U: use carefully for quick diagrams and definitions, then verify with textbook.
- IIT Bombay Electrical Machines lab manual: use for testing, lab procedure and measurement perspective.